Econometrics: Methods and Applications
Test Exercise 4

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Note: This PDF presents the final numerical results and interpretation for Test Exercise 4. All regressions, statistical tests and numerical computations were implemented in Python. The corresponding source code for this exercise is available in Module_5/scripts/test_ex_4.py, which is included in the GitHub repository referenced above. The code is extensively documented with comments explaining the methodology, implementation, and interpretation of the results. The repository also contains the training exercises and supporting work completed throughout this module.

a)

$$\log w = \beta_1 + \beta_2 educ + \beta_3 exper + \beta_4 exper^2 + \beta_5 smsa + \beta_6 south + \varepsilon$$

Results from OLS regression:

	b	SE	t-statistic
Constant	4.611	0.068	67.914
Education	0.082	0.003	23.315
Experience	0.084	0.007	12.377
Experience^2	-0.002	0	-6.8
Metropolitan	0.151	0.016	9.523
South	-0.175	0.015	-11.96

$R^2 \rightarrow 0.263$

The estimated $\beta_2 = 0.082$ implies that, holding experience and location variables constant, an additional year of schooling is associated with an approximately 8.2 % higher wage.

b)

Education and experience may be endogenous because unobserved factors such as ability, motivation, and ambition can affect both education and work experience, but also wage directly. Specifically, highly motivated people tend to pursue higher education, but also perform better in their job. Therefore, omitting motivation as a variable may cause the OLS estimate to overshoot the effect of education on wage. For experience, a similar argument can be made: More motivated or able individuals may pursue internships or full-time jobs sooner, gaining more experience. But they also perform better, which may also lead to increased wage. To conclude, both of these regressors may be correlated with the error term, where omitted variables like motivation may appear. However, even if the OLS estimates are biased due to endogeneity, they are still useful because they provide an initial estimate of the relationship between wage and the selected explanatory variables. They will also be useful for later comparison with the 2SLS method.

c)

Age and Age^2 can be motivated as instruments for experience and experience^2 because older individuals have had more time to accumulate work experience. Therefore, these instruments are likely to satisfy the correlation condition with the endogenous variable Experience.

Assuming Age and Age^2 are uncorrelated with the error term, they can be used as instruments for Experience and Experience^2

d)

The first stage regression, in which we explain Education with the exogenous variables and Age, Age^2, nearc, dadeduc, and momeduc as additional instruments, yields:

	b	SE	t-statistic
Constant	-5.652	3.976	-1.421
Metropolitan	0.53	0.102	5.217
South	-0.425	0.091	-4.667
Age	0.99	0.279	3.551
Age^2	-0.017	0.005	-3.518
Near College	0.265	0.099	2.67
Dad Educ	0.19	0.016	12.2
Mom Educ	0.235	0.017	13.773

$R^2 \rightarrow 0.247$

We now perform an F-test to test the joint significance of the gamma coefficients for the instruments used.

$\boxed{\text{F-statistic} = 145.511, \text{ with p-value} \approx 0}$ → Therefore, we reject the null hypothesis that $\gamma = 0$, meaning that the proposed instruments seem to be jointly significant predictors of schooling.

e)

Using 2SLS to correct for endogeneity of Education and Experience yields:

	b	SE	t-Statistic
Constant	4.417	0.115	38.27
Education	0.1	0.007	15.19
Experience	0.073	0.017	4.36
Experience^2	-0.002	0.001	-1.956
Metropolitan	0.135	0.017	8.047
South	-0.159	0.016	-10.136

$R^2 \rightarrow 0.251$

The 2SLS estimate of the return to schooling increases from 8.2% to approximately 10%, suggesting a larger causal effect of education on wages than implied by OLS.

The 2SLS estimate of the return to experience decreases from 8.4% to 7.3%, indicating a somewhat smaller effect of market experience after correcting for endogeneity.

f)

After performing the Sargan test to check whether the instruments of 2 are correlated with ε, we get:

$R^2 = 0.001$

Sargan-statistic $= nR^2 = 3.702$

p-value $= 0.157$

Since $0.157 > 0.05$, we fail to reject the null hypothesis of no correlation between the instruments and ε.

In this sense, the instruments used appear valid.